AI-Driven Lightweight and Trustworthy Federated Learning Framework for Decision-Making in Resource-Constrained Edge-IoT

Asadullah Tariq, *Member, IEEE*, Farag Sallabi, Mohamed Adel Serhani, Ezedin S. Barka, Tariq Qayyum, *Member, IEEE*, Irfanud Din, and Ikram Syed

***Abstract*— Artificial Intelligence (AI) and the Internet of Things (IoT) are transforming modern computing by enabling intelligent decision-making across distributed devices. Federated Learning (FL) facilitates decentralized model training while preserving data privacy, making it ideal for large-scale AI applications. However, FL faces a major challenge of high communication overhead due to frequent model updates between clients and the central server. Existing solutions, such as compressing gradients, struggle to balance efficiency and accuracy, particularly in resource-constrained edge environments. There is a growing need for trustworthy, communication-efficient FL solutions that ensure privacy, fairness, and security. To address this, we propose a lightweight novel Hierarchical Federated Learning (HFL) framework that integrates adaptive model pruning, quantization, and model communication and aggregation frequency optimization. First, we introduce a joint model pruning and quantization approach that dynamically adjusts pruning ratios and quantization levels, reducing communication costs while maintaining high accuracy. Second, we develop a fairness aware Stackelberg game-based model communication frequency optimization mechanism, where clients, edge servers, and the central server collaboratively determine optimal update frequencies to balance overhead and convergence speed. Third, we enhance privacy protection using Selective Homomorphic Encryption (SHE) and introduce a verifiable model trust assessment to ensure secure participation of edge devices. Extensive experiments validate our framework's effectiveness in optimizing communication efficiency, improving model accuracy, and ensuring fast convergence under diverse FL scenarios. The results demonstrate that our approach outperforms state-of-the-art methods in terms of reducing communication overhead, enhancing privacy, and achieving robust learning across varying network and computational constraints.**

Received 18 May 2025; revised 18 August 2025 and 6 October 2025; accepted 26 October 2025. Date of publication 3 November 2025; date of current version 8 December 2025. This work was supported by the Zayed Health Science Center under Grant 12R005. *(Corresponding author: Farag Sallabi.)*

Asadullah Tariq, Farag Sallabi, Ezedin S. Barka, and Tariq Qayyum are with the College of IT, United Arab Emirates University, Al Ain, United Arab Emirates (e-mail: 700039114@uaeu.ac.ae; f.sallabi@uaeu.ac.ae; Ebarka@uaeu.ac.ae; 700036923@uaeu.ac.ae).

Mohamed Adel Serhani is with the College of Computing and Informatics, University of Sharjah, Sharjah, United Arab Emirates (e-mail: mserhani@sharjah.ac.ae).

Irfanud Din is with New Uzbekistan University, Tashkent 100007, Uzbekistan (e-mail: irfan@newuu.uz).

Ikram Syed is with the Department of Information and Communication Engineering, Hankuk University of Foreign Studies, Yongin 17035, South Korea (e-mail: ikram@hufs.ac.kr).

Digital Object Identifier 10.1109/TCE.2025.3628207

***Index Terms*— Federated learning, internet of things, communication efficiency, responsible AI, privacy-preserving, trustworthy FL, interpretability.**

I. Introduction

ARTIFICIAL intelligence (AI) has rapidly evolved Over the past decade, driving breakthroughs in Machine Learning (ML) and Deep Learning (DL). Training highly accurate models for complex tasks like image recognition and natural language processing requires vast amounts of data. However, privacy concerns and communication limitations make centralized data storage impractical. Federated learning (FL) offers a decentralized solution, allowing AI models to be trained across multiple devices while keeping raw data local, enhancing privacy and reducing data transfer burdens [1].

A major challenge in FL is the high communication overhead caused by transmitting model updates between clients and the server. Modern deep neural networks (DNNs) contain millions of parameters, resulting in gradient sizes ranging from megabytes to gigabytes [2]. This becomes a bottleneck, especially for edge devices with limited bandwidth. Existing solutions focus on two strategies: (i) reducing model communication frequency by allowing multiple local updates before aggregation, which still results in large data transmissions, and (ii) applying gradient compression to minimize transmitted data per round. Communication efficiency in FL is significantly influenced by the neural network model's size. Smaller models enable smoother transmission between devices, reducing communication delays. While over-parameterization enhances generalization, it demands extensive resources, posing challenges in FL, where frequent local updates require substantial computation [3]. Neural network pruning addresses this by shrinking model size while maintaining functionality. Traditional pruning assesses parameter significance using a single dataset, but FL's non-IID data distribution complicates this process, causing overfitting or underfitting. Additionally, pruning at the client level may boost local performance but might not optimize the global model due to aggregation challenges.

One widely adopted compression technique is gradient quantization, where gradients are represented using a limited number of bits, determining the quantization levels. Lower quantization levels require fewer bits, reducing communication

1558-4127 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: BRAC UNIVERSITY. Downloaded on February 25,2026 at 08:15:36 UTC from IEEE Xplore. Restrictions apply.

overhead but introducing quantization errors, which can slow convergence [4]. Conversely, higher quantization resolutions reduce errors but increase data transmission, leading to longer overall training durations. Thus, selecting an appropriate quantization level is crucial for optimizing training time [5]. Existing quantization approaches generally use fixed, predefined settings throughout training. However, different FL tasks exhibit diverse characteristics, such as convergence speed, communication constraints, and network variability, making static quantization less effective. Our observations indicate that gradient values fluctuate significantly during training, suggesting that dynamically adjusting quantization levels across rounds could enhance efficiency and overall performance [6].

Efficient communication frequency optimization of model updates is critical in FL to balance model convergence and system efficiency. Excessive updates increase communication overhead, while infrequent transmissions risk model divergence [7]. Adaptive communication strategies help reduce bandwidth consumption while maintaining model accuracy and training stability, ensuring effective learning in decentralized environments. This encryption technique enables computations on encrypted data without requiring decryption, preserving confidentiality throughout the learning process. By preventing data exposure and mitigating security risks, Homomorphic Encryption (HE) enhances privacy protection, reducing vulnerabilities to breaches and adversarial attack. In the context of FL, it enables model updates to remain encrypted during aggregation, preserving data privacy while still allowing accurate global model training. Despite significant advancements in communication-efficient FL, several challenges remain unaddressed for further performance enhancement. First, the impact of model pruning on learning performance, particularly the errors introduced by removing parameters, requires a thorough mathematical analysis. Additionally, inefficient devices with limited computational power or poor network conditions can introduce delays and aggregation inconsistencies, necessitating their exclusion to minimize communication overhead. Moreover, optimizing the pruning ratio, device selection, and wireless resource allocation is crucial for improving overall FL efficiency. Hierarchical FL (HFL) extends traditional FL by introducing intermediate edge servers that aggregate client updates before sending them to the central server. These servers perform partial aggregation before forwarding results to the central server. This reduces communication overhead, enhances scalability, and improves robustness in large or resource-constrained networks. The architecture of HFL is shown in Fig. 1. It enables adaptive model pruning across layers, optimizing accuracy and resource use. The key contributions of this paper are summarized as follows:

1) An EdgeFL-based novel joint model pruning and quantization framework that dynamically adjusts pruning ratios and quantization levels, optimizing communication efficiency, reducing overhead, and ensuring high model accuracy while maintaining fast convergence in FL.
2) A novel FL model communication and aggregation frequency optimization model using the Stackelberg game, dynamically balancing communication overhead and model convergence. This approach enhances efficiency by optimizing update frequency based on hierarchical decision-making in FL.
3) To enhance privacy, we integrate Selective Homomorphic Encryption (SHE) for secure computations and propose a verifiable model to assess the trustworthiness of edge devices, ensuring data confidentiality and reliable participation in FL.
4) Extensive experimental evaluations demonstrate the superior performance of our proposed framework in terms of communication efficiency, model accuracy, privacy preservation, and convergence speed, highlighting its effectiveness in optimizing FL under diverse network and computational constraints.

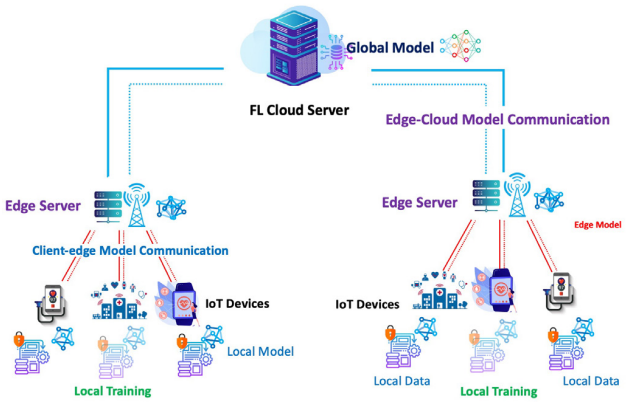


Fig. 1. Hierarchical FL Architecture.

The remainder of this paper is organized as follows: Section II discusses related work, providing an overview of existing approaches and their limitations. Section III introduces the proposed methodology, detailing our novel framework for communication-efficient and privacy-preserving FL. Section IV presents the experimental evaluations and results. Finally, Section V concludes the paper and outlines potential directions for future research.

II. Related Works

Some studies [8], [9] have comprehensively explored trustworthy and communication-efficient FL, including detailed discussions on quantization and pruning. In this section, we discuss recent state-of-the-art research related to communication efficient pruning and quantization in FL and the motivation behind our proposed work.

A. Pruning

The research works [10], [11], [12], [13], [14] introduced adaptive and distributed pruning strategies for communication efficient FL. PruneFL [10] dynamically adjusting model size to minimize computation and communication overhead. It follows progressive pruning that continuously refines parameter selection, maintaining a compact yet effective model that achieves near-original accuracy. AMP-FL in [11] dynamically prunes the global model to generate sub-models according to devices' computing and communication capabilities. It compensates for pruned gradients

Authorized licensed use limited to: BRAC UNIVERSITY. Downloaded on February 25,2026 at 08:15:36 UTC from IEEE Xplore. Restrictions apply.

using historical updates, leverages an Age of Information (AoI) metric for optimized device scheduling, and formulates resource block allocation as a tractable optimization problem, ensuring efficient FL convergence. A threshold-based strategy is used in [12] to removes straggling devices to mitigate pruning-induced aggregation loss and reduce communication overhead., while analytical solutions guide optimal pruning and resource allocation. VFed-AMP [13] employs a low-complexity iterative algorithm that adjusts pruning parameters dynamically, minimizing computation and communication overhead while adapting to IoV mobility and network fluctuations for optimal FL performance. FedPE in [14] dynamically adjust model size based on real-time accuracy variations. It integrates error compensation to preserve pruned information and an accuracy-driven aggregation rule for fair model updates. A Pareto-based hyperparameter optimization algorithm ensures an optimal trade-off between accuracy and efficiency, improving FL robustness. Authors in [50] proposed an explainable, data-quality–driven sample selection mechanism, which represents a promising approach toward achieving communication-efficient federated learning. FedPAGE [16] employs a pruning rate learning approach, dynamically adjusting pruning rates based on worker capacity, eliminating stragglers. In FedMP [15], A Multi-Armed Bandit (MAB)-based online learning algorithm dynamically adjusts pruning ratios. Additionally, a Residual Recovery Synchronous Parallel (R2SP) scheme ensures stable model aggregation.

DF2-MPC in [17], introduced personalized pruning strategy for adaptive compression based on communication capacity and neighbors, reducing transmission delay and mitigating collusion risks. Additionally, an adaptive neighbor selection scheme optimizes connections under resource constraints by considering data heterogeneity, model quality, and consensus. Moreover, the proposed FL framework in [18] employs partial model pruning and personalization, dividing the model into a global shared part and a personalized fine-tuned part to adapt dynamically during training. A server-side similarity-based pruning approach in [19] employs global-local model similarity to identify and prune less significant parameters. It follows an Aggregation-Pruning-Aggregation sequence, ensuring models incorporate the latest global updates before pruning. By leveraging the Lottery Ticket Hypothesis, FedADP in [20] extracts high-accuracy subnetworks for efficient training. It only transfer subnetwork parameters instead of the full model. In addition, Fed-LTP [21] integrates Lottery Ticket Hypothesis (LTH) and zero-concentrated Differential Privacy (zCDP) for efficient, privacy-preserving FL. It employs weight-based and iterative pruning on the server to reduce model size and sparse-to-sparse training on clients for resource efficiency. A zCDP-based privacy analysis optimizes the utility-privacy trade-off while minimizing overhead. GHPPFL [22] proposes a FL method combining gradient compression to reduce data transfer with homomorphic encryption for enhanced gradient privacy.

FedLP [23] employs a layer-wise pruning mechanism during local training and federated updates. The framework optimizes pruning granularity per layer, minimizing communication and computation overhead while maintaining FL convergence and robustness. FedGroup-Prune [24] follows a two-stage pruning process: (1) Local sparsity training via Group Lasso generates a sparse matrix, and (2) Neuron group pruning removes low-weight connections, reducing model complexity. PrVFL [25] employs a zero-knowledge range proof protocol to verify model pruning correctness without exposing parameter details. The global residual model allows dynamic pruning ratio adjustments, ensuring efficient training while maintaining verification integrity. GWEP in [26] reduces DNN size through joint quantization and pruning, lowering computational, memory, and communication demands.

The proposed adaptive model pruning framework in [27] for HFL models formulates a joint optimization problem to maximize convergence under latency constraints. By decoupling the optimization problem into subproblems and applying Karush–Kuhn–Tucker (KKT) conditions, it derives closed-form solutions for optimal pruning ratio and wireless resource allocation, ensuring efficient training in dynamic wireless environments. A joint model pruning and wireless resource allocation framework for HFL in wireless networks to reduce computation and communication latencyb [28]. It mathematically analyzes the impact of pruning on latency, derives the convergence rate with pruning, and formulates an optimization problem. The proposed PHFL framework [29] optimizes model pruning, CPU frequency, and transmission power in HFL under wireless HetNet constraints. It first derives a convergence bound capturing pruning and communication impacts. Then, a joint optimization problem is formulated to maximize convergence. Due to non-convexity, successive convex approximation (SCA) is applied to solve the relaxed convex problem, ensuring efficient resource allocation and improved learning performance.

B. Quantization

Adaptive Quantized Gradient (AQG) [30] adjusts quantization based on local gradients to reduce transmissions and mitigates client dropouts with augmented AQG. DAdaQuant in [31], combines time- and client-adaptive quantization builds on the QSGD fixed-point quantizer. Itdynamically adjusts quantization levels based on client contribution and training progress. AdaGQ [2] adjusts resolution based on gradient norm variations, and heterogeneous quantization assigns lower resolution to slower clients and higher resolution to faster ones. NQFL in [32], a nonuniform quantization scheme leveraging the Lloyd-Max algorithm. This approach normalizes gradients to use a shared quantizer, reducing computational and communication overhead. TLAQC [33] introduces a two-layer accumulated quantized compression algorithm, combining a revised quantization method (RQSGD) with communication sparsification. RQSGD applies zero-value correction to reduce ineffective quantization, while an adaptive threshold determines gradient sparsification. FedSQ in [34] combines sparsity and quantization that selectively uploads parameters with significant gradient changes while retaining previous-round values for those with minimal changes. An error compensation mechanism corrects compression loss, ensuring consistency between current and previous model parameters.

Authorized licensed use limited to: BRAC UNIVERSITY. Downloaded on February 25,2026 at 08:15:36 UTC from IEEE Xplore. Restrictions apply.

SOBAA [35] combines layer-wise sparsification, one-bit quantization, power control, and an error-feedback mechanism to improve convergence. The method optimizes the compression ratio and power control to minimize errors, accelerating convergence. QuAsyncFL [36] incorporates gradient quantization and weight aggregation while analyzing convergence speed under different quantization levels. The framework in [37] incorporates stochastic quantization that quantizes local gradients into discrete values, reducing communication load. A quantization-interval parameter controls the noise level, balancing gradient sparsity and protection. Stochastic quantization in [38] divides gradient intervals uniformly, encoding them into bits, balancing resource efficiency and precision.

The research works in [39], [40], [42], and [43], introduces methods to dynamically adjust quantization levels for model updates in FL. Unlike traditional methods, AdaQuantFL [39] bounds error based on the number of bits communicated rather than training rounds. This approach builds on [40], which adapts local updates and communication frequency. FedDQ in [41] adapting quantization levels based on shrinking model updates during training, FedDQ reduces communication volume and rounds. The proposed work in [42] quantizes local model parameters by analyzing quantization errors and dynamically adjusts quantization bits and resources. It uses Lagrange dual decomposition for optimization. FedMPQ [43] introduces mixed-precision quantization (MPQ) by adaptively training quantized local models within bit-width constraints. The server aggregates de-quantized models into a global model and adjusts layer bit-widths for each client based on resource availability and prior rounds.

FedQNN in [45], uniquely integrates ultralow-bitwidth quantization, enabling clients to perform lightweight fixed-point computations with minimal power usage. Authors in [46] proposes a non-uniform quantization for DFL using the Lloyd-Max algorithm to minimize distortion and improve convergence. It introduces a doubly-adaptive approach, adjusting quantization based on gradient distribution. Hier-Local-QSGD [47] is a hierarchical FL framework with model quantization. Clients send quantized updates to edge servers, and edge servers further send quantized updates to the cloud. The methodology optimizes the client-edge aggregation interval and the cloud-edge communication strategy. FEDHQ and FEDHQ+ in [48] assigns dynamic weights to clients based on heterogeneous quantization errors and instantaneous quantization error feedback from clients respectively. FEDHBAA [51] combines heterogeneous quantization bit allocation and server aggregation to enhance FL by dynamically allocating quantization bits based on client errors. Reference [53] introduced a distributed DRL-based approach for quantization level allocation, aimed at optimizing long-term rewards based on total training time and quantization efficiency.

C. Model Communication and Aggregation Frequency Optimization

With limited prior work on communication frequency of model updates in HFL, the research work in [54] addresses the gap by jointly optimizing aggregation frequency and edge association via convex optimization and reinforcement learning, minimizing energy while preserving convergence under non-IID data, and outperforming benchmarks significantly. Moreover, another work in [55] introduced RAF, a resource based method to set optimal aggregation frequencies per edge device reducing waiting time handling weak synchronization achieving faster convergence and improved accuracy in hierarchical federated learning. The Lazily Aggregated Quantized (LAQ) in [58] involves adaptive communication of quantized gradients, where gradients are selectively quantized and reused, reducing the need for frequent updates. AQUILA [59] improves FL efficiency through adaptive quantization and selective device updates. It adjusts quantization dynamically, employs deterministic rounding for compression, reduces transmissions using a skipping rule, and optimizes aggregation to minimize model deviation. The research work in [52] presents a Trust-Based Dynamic Node Selection and Aggregation Frequency Optimization method, employing a Deep Q-Network (DQN) to address this challenge. Fed-PEMC [49] combines model compression with local differential privacy to enhance communication efficiency, reduce model size, and protect against inference attacks. Reinforcement learning and customized label sampling are also introduced to accelerate training. Several works have explored adaptive quantization and pruning, but they often lack privacy preservation and overlook communication frequency optimization in FL. While some approaches address these challenges separately, integrating all three model adaptive pruning, quantization, and privacy along with communication frequency optimization remains underexplored. The comparative analysis of related works with our proposed work is presented in Table I.

III. Proposed Methodology

In this paper, we have implemented a structured approach integrating feature dropout, adaptive pruning, and adaptive quantization to enhance the efficiency of FL. Feature dropout was applied before local training to reduce communication overhead by eliminating less significant features while ensuring adaptive selection to retain critical information. After several training rounds, we introduced structured adaptive pruning to optimize model complexity by gradually removing redundant neurons or layers. This step was carefully coordinated across clients to maintain consistency in global model aggregation and prevent accuracy degradation. Finally, adaptive quantization was applied before transmitting model updates to the edge and central server. This approach minimized transmission costs while preserving model accuracy. The detailed representation of our proposed framework in presented in Fig. 2.

A. Problem Formulation

In this work, our HFL system consists of three key components: clients, edge servers, and a central server. The clients, denoted as $\mathcal{C} = \{C_1, \ldots, C_N\}$, each possess a local dataset $\mathcal{D}_i$ that follows a distribution $p_i(x, y)$. These clients are grouped and managed by edge servers, represented as $\mathcal{E} = \{E_1, \ldots, E_E\}$, where each edge server E_e oversees a subset of

Authorized licensed use limited to: BRAC UNIVERSITY. Downloaded on February 25,2026 at 08:15:36 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPARATIVE ANALYSIS OF RELATED WORKS

Type	Related Work	FL	Communication Efficient	Adaptive	Non-Adaptive	Personalized	Hierarchical FL	Privacy	Verifiable	Commun. Frequency Optimization	Secure Aggregation
Pruning	[10], [11], [12], [13]	✓	✓	✓	✗	✗	✗	✗	✗	✗	✗
	[14], [16], [19], [23]	✓	✓	✗	✓	✗	✗	✗	✗	✗	✓
	[17], [18]	✓	✓	✗	✓	✓	✗	✓	✗	✗	✗
	[21], [25]	✓	✓	✗	✓	✗	✗	✓	✓	✗	✗
	[27], [28], [29]	✓	✓	✗	✗	✗	✓	✗	✗	✗	✗
Quantization	[30], [31], [2], [32], [33]	✓	✓	✓	✗	✗	✗	✗	✗	✗	✗
	[39], [40], [42], [43], [44]	✓	✓	✓	✗	✗	✗	✗	✗	✗	✗
	[51], [53]	✓	✓	✗	✓	✗	✗	✓	✓	✗	✓
	[47], [48]	✓	✓	✗	✓	✗	✓	✗	✗	✗	✗
	[58], [59]	✓	✓	✗	✓	✗	✗	✗	✓	✓	✗
Commun. Frequency Optimization	[54]	✓	✓	✗	✓	✗	✓	✓	✗	✓	✗
	[55]	✓	✓	✗	✓	✗	✓	✓	✗	✓	✗
	[58], [59]	✓	✓	✗	✓	✗	✗	✗	✓	✓	✗
This Work		✓	✓	✓	✗	✓	✓	✓	✓	✓	✓

clients $\mathcal{S}_e \subseteq \mathcal{C}$. At the highest level, a central server performs the global model aggregation by coordinating updates from the edge servers.

The global objective of the system is to minimize the loss function $\mathcal{L}(W)$, defined as:

$$\min_W \mathcal{L}(W) = \frac{1}{N} \sum_{i=1}^{N} \mathbb{E}_{(x,y)\sim p_i} \left[\mathcal{L}(W, x, y) \right], \tag{1}$$

where W is the global model, $\mathcal{L}(W, x, y)$ is the local loss function for data point (x, y), and $p_i(x, y)$ represents the data distribution for client C_i.

B. Adaptive Feature-Wise Dropout

Adaptive Feature-Wise Dropout (AFD) regularizes a model by assigning feature-specific dropout probabilities. For a dataset $\mathbf{X} \in \mathbb{R}^{n\times d}$, the importance of feature j is its standard deviation $\sigma_j = \sqrt{1n \sum_{i=1}^{n} (x_{ij} - \mu_j)^2}$, where $\mu_j = 1n \sum_{i=1}^{n} x_{ij}$. Normalizing gives $\tilde{\sigma}_j = \sigma_j / \sum_{k=1}^{d} \sigma_k$, and the dropout probability becomes $p_j = 1 - \tilde{\sigma}_j$. Each mask element is drawn as $m_j \sim Bernoulli(1 - p_j)$ and applied to an input $\mathbf{x}$ via $\tilde{\mathbf{x}} = \mathbf{m} \odot \mathbf{x}$. Because $\mathbb{E}[m_j] = \tilde{\sigma}_j$, the expected number of retained features is $\sum_{j=1}^{d} \tilde{\sigma}_j = 1$, giving an expected retained-feature fraction $\mathbb{E}[f_r] = 1/d$ and a communication saving of $1 - \mathbb{E}[f_r] = 1 - 1/d$. For a linear layer with weights $\mathbf{w}$, the usual output $y = \mathbf{w}^\top \mathbf{x}$ becomes $\tilde{y} = \mathbf{w}^\top (\mathbf{m} \odot \mathbf{x})$, whose expectation is $\mathbb{E}[\tilde{y}] = \mathbf{w}^\top (\mathbb{E}[\mathbf{m}] \odot \mathbf{x})$ with $\mathbb{E}[\mathbf{m}] = [\tilde{\sigma}_1, \ldots, \tilde{\sigma}_d]^\top$. Hence AFD preferentially keeps informative, high-variance dimensions while reducing computation and bandwidth.

C. Adaptive Pruning in HFL

While Adaptive Feature-Wise Dropout selectively deactivates less important features during training, model pruning extends this idea by permanently removing redundant parameters, significantly reducing model complexity. In FL, transmitting large model updates across multiple clients and edge servers poses a major scalability challenge. Pruning helps mitigate this by reducing the number of parameters transmitted, thereby improving both computational and communication efficiency. In this section, we introduce *Adaptive Pruning in HFL*, a framework that integrates structured parameter pruning into a hierarchical FL system. By leveraging the hierarchical aggregation structure, pruning decisions are refined at both client and edge-server levels before final global model aggregation.

Unlike conventional pruning that applies a fixed threshold to eliminate weights, our adaptive pruning framework dynamically determines which parameters to remove based on their deviation during training. This is inspired by the concept that parameters with minimal changes contribute less to model learning and can be pruned without significantly affecting convergence. For each client $i \in \mathcal{C}_e$, let $\mathbf{W}_i \in \mathbb{R}^d$ represent the local model parameters. The importance of parameter m is measured by its deviation over a training phase:

$$D_{i,m} = \left(W_{i,m}^{\text{initial}} - W_{i,m}^{\text{final}} \right)^2, \tag{2}$$

where $W_{i,m}^{\text{initial}}$ and $W_{i,m}^{\text{final}}$ denote the parameter values at the beginning and end of a training phase, respectively. Since individual deviations may be noisy, edge servers aggregate deviation values across all their clients:

$$D_{e,m} = \frac{1}{|\mathcal{C}_e|} \sum_{i \in \mathcal{C}_e} D_{i,m}. \tag{3}$$

Authorized licensed use limited to: BRAC UNIVERSITY. Downloaded on February 25,2026 at 08:15:36 UTC from IEEE Xplore. Restrictions apply.

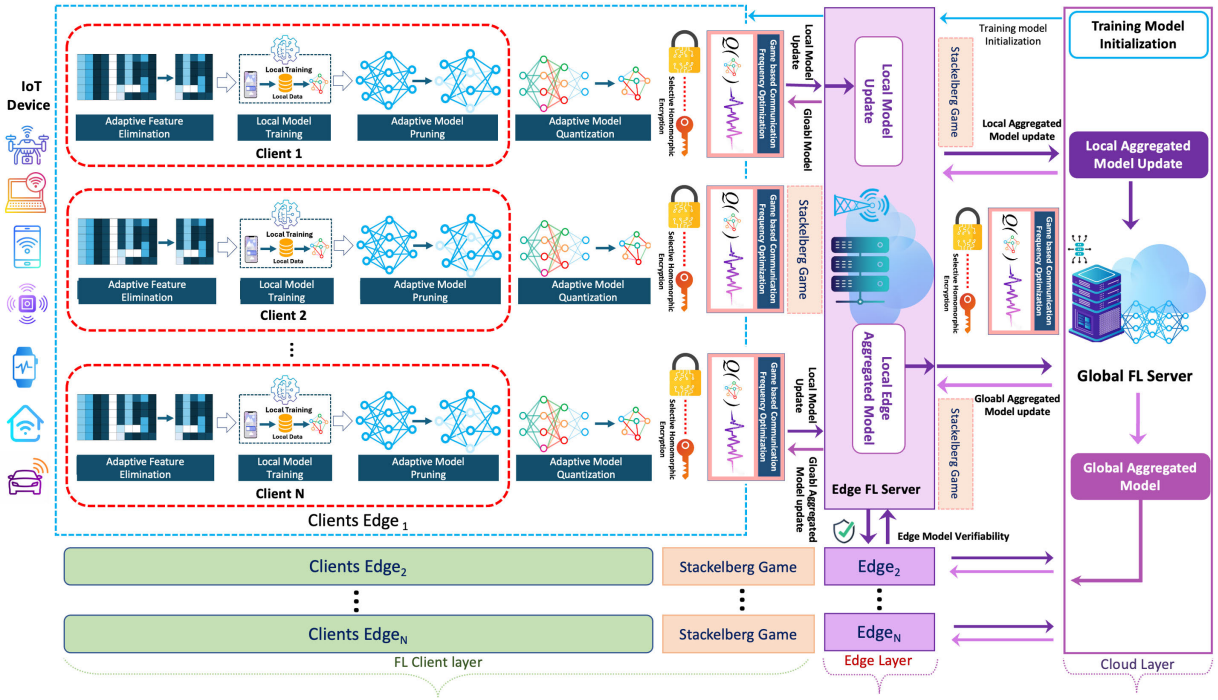


Fig. 2. An overview of the proposed communication efficient HFL system model, integrating Adaptive Feature-Wise Dropout, Adaptive Pruning, and Dynamic Quantization, supported by Game-Theoretic Model Communication & Aggregation Frequency Optimization empowered by privacy-preserving mechanism.

These edge-level deviations are then forwarded to the central server, which performs global aggregation:

$$D_{g,m} = \frac{1}{|\mathcal{E}|} \sum_{e \in \mathcal{E}} D_{e,m}. \tag{4}$$

A pruning mask $\mathbf{M} \in \{0,1\}^d$ is then generated based on a global threshold T:

$$M_m = \mathbb{I}(D_{g,m} > T), \tag{5}$$

where $\mathbb{I}(\cdot)$ is the indicator function. Parameters with $D_{g,m} \leq T$ are pruned (i.e., set to zero), while others are retained. Once the pruning mask is generated, each client applies it to its local model before transmission:

$$\mathbf{W}_i^{\text{pruned}} = \mathbf{W}_i \odot \mathbf{M}, \tag{6}$$

where $\odot$ denotes element-wise multiplication. Edge servers then aggregate the pruned updates from their clients:

$$\mathbf{W}_e = \frac{1}{|\mathcal{C}_e|} \sum_{i \in \mathcal{C}_e} \mathbf{W}_i^{\text{pruned}}. \tag{7}$$

Even though pruning produces sparse local updates, we apply weight averaging on the retained parameters to ensure consistency in global aggregation and maintain convergence stability. Similarly, the central server aggregates the updates from all edge servers:

$$\mathbf{W}_g = \frac{1}{|\mathcal{E}|} \sum_{e \in \mathcal{E}} \mathbf{W}_e. \tag{8}$$

This ensures that only the most critical parameters are retained in the global model. We analyze the convergence properties of the pruned model under the assumption of Lipschitz continuity and bounded gradients. The global loss function is defined as:

$$L(\mathbf{W}) = \frac{1}{|\mathcal{E}|} \sum_{e \in \mathcal{E}} \frac{1}{|\mathcal{C}_e|} \sum_{i \in \mathcal{C}_e} L_i(\mathbf{W}). \tag{9}$$

Let $\nabla L(\mathbf{W})$ be the gradient of the global loss. The pruned gradient, denoted by $\nabla L^{\text{pruned}}(\mathbf{W})$, can be expressed as:

$$\nabla L^{\text{pruned}}(\mathbf{W}) = \nabla L(\mathbf{W}) \odot \mathbf{M}. \tag{10}$$

The pruning error is bounded by following equation, where ϵ is a function of the pruning threshold T and the parameter deviation distribution. This ensures that pruning does not introduce excessive error, maintaining convergence stability.

$$\|\nabla L(\mathbf{W}) - \nabla L^{\text{pruned}}(\mathbf{W})\|_2 \leq \epsilon, \tag{11}$$

Assuming that $L(\mathbf{W})$ is μ-strongly convex and $\nabla L(\mathbf{W})$ is L-Lipschitz continuous, the pruned model converges to the optimal global model $\mathbf{W}^*$ at the rate:

$$L(\mathbf{W}_g^{(t)}) - L(\mathbf{W}^*) \leq \frac{L}{2t} \|\mathbf{W}_g^{(0)} - \mathbf{W}^*\|_2^2. \tag{12}$$

This indicates that despite parameter reduction, the model maintains convergence properties under standard assumptions. By pruning parameters at both client and edge levels, the total number of transmitted parameters is significantly reduced. The fraction of retained parameters is given by:

$$f = \frac{\|\mathbf{M}\|_0}{d}, \tag{13}$$

where $\|\mathbf{M}\|_0$ represents the number of nonzero elements in $\mathbf{M}$. The corresponding communication reduction factor is $1 - f$. This reduction is crucial for improving the scalability of FL, especially in environments with constrained bandwidth. Algorithm 1 summarizes the working of adaptive pruning in our proposed work.

Authorized licensed use limited to: BRAC UNIVERSITY. Downloaded on February 25,2026 at 08:15:36 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 Adaptive Pruning in Proposed HFL Framework

Input: Local models $\mathbf{W}_i$, pruning threshold T
Output: Pruned global model $\mathbf{W}_g$
for each client $i \in \mathcal{C}_e$ **do**
  Compute parameter deviation $D_{i,m} = (W_{i,m}^{\text{init}} - W_{i,m}^{\text{final}})^2$
  Send $D_{i,m}$ to edge server
end for
for each edge server $e \in \mathcal{E}$ **do**
  Aggregate deviations: $D_{e,m} = \frac{1}{|\mathcal{C}_e|} \sum_i D_{i,m}$
  Send to central server
end for
Compute global deviation: $D_{g,m} = \frac{1}{|\mathcal{E}|} \sum_e D_{e,m}$
Generate pruning mask: $M_m = \mathbb{I}(D_{g,m} > T)$
for each client i **do**
  Apply mask: $\mathbf{W}_i^{\text{pruned}} = \mathbf{W}_i \odot \mathbf{M}$
  Send to edge server
end for
Aggregate at edge and central servers:

$$\mathbf{W}_e = \frac{1}{|\mathcal{C}_e|} \sum_i \mathbf{W}_i^{\text{pruned}}, \quad \mathbf{W}_g = \frac{1}{|\mathcal{E}|} \sum_e \mathbf{W}_e$$

Return: Pruned global model $\mathbf{W}_g$

D. Adaptive Quantization

While adaptive pruning significantly reduces the number of transmitted parameters by removing less important weights, further efficiency gains can be achieved through quantization. In this part of the section, we introduce Adaptive Quantization strategy, where quantization levels are adaptively adjusted based on gradient characteristics, ensuring an optimal trade-off between compression and accuracy.

Consider a client k with model parameters $\mathbf{w}_k$. Instead of transmitting full-precision values, the client applies a quantization function $Q(\cdot; \alpha_k)$, where α_k represents the quantization level:

$$\mathbf{w}_k^Q = Q(\mathbf{w}_k; \alpha_k). \tag{14}$$

The quantization level α_k is dynamically updated based on the gradient distribution, ensuring that significant updates are preserved while smaller updates are aggressively compressed:

$$\alpha_k = \frac{\|\nabla F_k(\mathbf{w}_k)\|_2}{\|\nabla F_k(\mathbf{w}_k)\|_\infty}. \tag{15}$$

This formulation ensures that the quantization level is adapted to the scale of model updates, preventing excessive precision loss for large gradient values while still compressing redundant information. Quantization introduces a small approximation error, denoted as ϵ_k, which is bounded by:

$$\|\mathbf{w}_k - \mathbf{w}_k^Q\|_2 \leq \epsilon_k, \quad \text{where} \quad \epsilon_k = \frac{\alpha_k}{2}. \tag{16}$$

To mitigate the impact of this error on model convergence, we introduce an *error feedback mechanism*, which compensates for quantization loss in subsequent updates. The adjusted model update rule at client k is:

where β is the feedback coefficient controlling the correction intensity. This mechanism ensures that quantization errors do not accumulate over time, maintaining model stability. To eliminate quantization error over time, we maintain a residual $r^t = w^t - Q(w^t)$ and add βr^t in Eq. (18). With diminishing η_t and bounded quantization bias $\|w - Q(w)\| \leq \epsilon_t$, error-feedback prevents error accumulation and yields a convergence rate of $O(1/T) + O\left(\sum_k \bar{\epsilon}_k / T\right)$. When $\|\nabla F_k(w)\|$ is large, α_k increases the bit-width (Eq. 16), thereby reducing ϵ_t in high-error phases and preserving stability.

1) Convergence Analysis: We analyze the convergence of dynamic quantization under the assumptions of L-Lipschitz smoothness and μ-strong convexity. The expected gap between the final model $\mathbf{w}_T$ and the optimal model $\mathbf{w}^*$ is given by:

$$\mathbb{E}[F(\mathbf{w}_T) - F(\mathbf{w}^*)] \leq \mathcal{O}\left(\frac{1}{\mu T} + \frac{\sum_{k=1}^K \epsilon_k}{\sqrt{T}}\right), \tag{17}$$

where T is the total number of training iterations. The first term represents the standard convergence rate in a convex setting, while the second term quantifies the impact of quantization errors. By carefully tuning α_k and incorporating error feedback, the overall impact of quantization error can be minimized.

2) Final Communication Cost Reduction: Combining pruning and quantization, the fraction of retained parameters is f, and each retained parameter is encoded using b bits instead of the original 32-bit floating-point representation. The total communication cost reduction factor is:

$$C_{\text{reduction}} = (1 - f) + f \cdot \frac{b}{32}. \tag{18}$$

By carefully tuning f (pruning level) and b (quantization bit-width), an optimal trade-off between efficiency and model accuracy can be achieved.

E. Model Communication and Aggregation Frequency Optimization

We performed empirical testing with various communication frequencies, as shown in Fig. 3. The results clearly indicate that more frequent communication leads to higher accuracy. However, this also introduces significant communication and computational overhead. Therefore, designing an efficient model communication and aggregation frequency optimization strategy is crucial. In HFL, model communication frequency optimization is crucial for reducing communication costs while ensuring efficient model convergence. It can be framed as a game-theoretic problem in which different entities (clients, edge servers, and the central server) interact and make decisions to maximize their utilities. In this section, we propose a Stackelberg game-based framework for optimizing the dynamic communication aggregation frequency across hierarchical levels. We consider a HFL system with N clients, each with a local dataset $D_i \sim p_i(x, y)$, E edge servers $E = \{E_1, E_2, \ldots, E_E\}$, and a central server that sets a global policy parameter δ.

In our system, Communication occurs in three stages:

1) *Client-to-Edge Communication:* Clients communicate their model updates to edge servers at a frequency f_i, where $f_i \in [0, 1]$ represents the proportion of local updates communicated to the edge server.

Authorized licensed use limited to: BRAC UNIVERSITY. Downloaded on February 25,2026 at 08:15:36 UTC from IEEE Xplore. Restrictions apply.

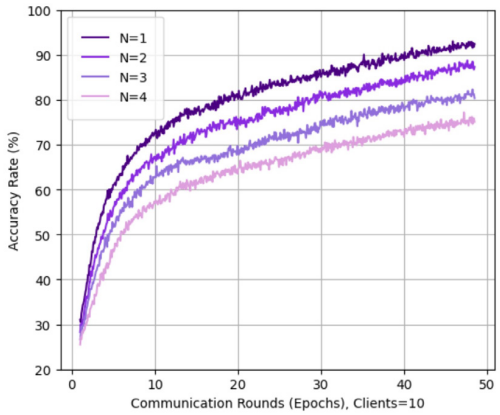


Fig. 3. Empirical analysis of accuracy over different communication frequencies on simple FedAvg settings in FL on MNIST dataset.

2) *Edge-to-Central Communication:* Edge servers aggregate updates and communicate with the central server at a frequency f_e, where $f_e \in [0, 1]$.
3) *Global Aggregation:* The central server aggregates all updates to update the global model.

Each client i decides its model updates' communication frequency f_i. Each edge server e determines f_e, considering the aggregated behavior of its clients and the central server's policy δ. The central server optimizes the global policy parameter δ to influence edge server frequencies f_e. The goal is to find an optimal hierarchical communication strategy $\{f_i, f_e\}$ and policy δ that minimize the overall cost while maintaining model accuracy and convergence. The working of communication frequency optimization module is summarized in Algorithm 2.

$$\mathbf{w}_k^{t+1} = \mathbf{w}_k^t - \eta_t \nabla F_k(\mathbf{w}_k^t) + \beta(\mathbf{w}_k^t - \mathbf{w}_k^{Q,t}), \tag{19}$$

1) Game-Theoretic Framework: We model the problem as a Stackelberg game:

- *Leader:* The central server sets the global policy δ, anticipating the response of edge servers and clients.
- *Followers:* Edge servers and clients react to δ while optimizing their utilities.

The utility of client i depends on communication costc $C_i(f_i)$ where increasing f_i raises communication costs and accuracy loss $\epsilon_p(f_i) + \epsilon_q(f_i)$ where Lower f_i increases pruning (ϵ_p) and quantization (ϵ_q) errors. The utility function of client i is:

$$U_i(f_i) = -C_i(f_i) - \alpha_p \epsilon_p(f_i) - \alpha_q \epsilon_q(f_i), \tag{20}$$

where $\alpha_p, \alpha_q > 0$ weight the accuracy losses. The utility of edge server e aggregates the behavior of its client set $\mathcal{S}_e$ where the aggregated communication cost is:

$$C_e(f_e) = \sum_{i \in \mathcal{S}_e} C_i(f_i) + \beta_e C_e^{\text{inter}}(f_e), \tag{21}$$

where $C_e^{\text{inter}}(f_e)$ is the edge-to-central communication cost. Accuracy loss here is Pruning error $\epsilon_p(f_e)$ due to edge-level compression. The utility function of edge server e is:

$$U_e(f_e) = -C_e(f_e) - \gamma_e \epsilon_p(f_e), \tag{22}$$

Algorithm 2 Optimized Model Communication and Aggregation Frequency via Stackelberg Game

Init: Global policy δ, client freq. f_i, server freq. f_e
while not converged **do**
  for each client i **do**
    Update f_i: $f_i^{(t+1)} = f_i^{(t)} + \eta_i \frac{\partial U_i}{\partial f_i}$
  end for
  for each server e **do**
    Update f_e: $f_e^{(t+1)} = f_e^{(t)} + \eta_e \frac{\partial U_e}{\partial f_e}$
  end for
  Update δ: $\delta^{(t+1)} = \delta^{(t)} + \eta_c \frac{\partial U_c}{\partial \delta}$
end while
Output: Optimized $\{f_i^*, f_e^*, \delta^*\}$

where $\beta_e, \gamma_e > 0$ are weighting parameters. The central server's utility aggregates global costs and accuracy:

$$U_c(\delta) = -\sum_{e=1}^{E} C_e(f_e(\delta)) - \delta \cdot \text{Global Error}(\{f_e(\delta)\}), \tag{23}$$

where $\delta > 0$ balances cost and accuracy, and $f_e(\delta)$ denotes edge frequencies influenced by δ.

2) Stackelberg Game Setup: The Stackelberg game is defined as:

$$G = (P, S, U), \tag{24}$$

where P are the Players (clients, edge servers, central server), S is the Strategy spaces ($f_i \in [0, 1]$, $f_e \in [0, 1]$, $\delta \in \mathbb{R}^+$), and U is the Utility functions (U_i, U_e, U_c).

In HFL framework, clients are considered the lowest-level decision-makers. Given a fixed global policy parameter δ set by the central server and an aggregation frequency f_e determined by edge servers, each client i aims to optimize its communication frequency f_i to maximize its utility function. Mathematically, each client i solves the following optimization problem:

$$\forall i, \quad f_i^* = \arg \max_{f_i \in [0,1]} U_i(f_i). \tag{25}$$

Since $U_i(f_i)$ is a differentiable function of f_i, the necessary first-order optimality condition for a maximum is obtained by setting its derivative to zero:

$$\frac{\partial U_i}{\partial f_i} = -\frac{\partial C_i}{\partial f_i} - \alpha_p \frac{\partial \epsilon_p}{\partial f_i} - \alpha_q \frac{\partial \epsilon_q}{\partial f_i} = 0. \tag{26}$$

Here, $\frac{\partial C_i}{\partial f_i}$ represents the marginal communication cost increase due to higher model communication frequency, while $\frac{\partial \epsilon_p}{\partial f_i}$ and $\frac{\partial \epsilon_q}{\partial f_i}$ quantify the impact of pruning and quantization errors, respectively.

At the intermediate level, edge servers play a crucial role in aggregating client updates and forwarding them to the central server. Each edge server e aggregates the local updates from its assigned clients $\mathcal{S}_e$ and determines its own communication frequency f_e. Given the central server's global policy δ, each edge server seeks to optimize its frequency to maximize its utility function. Formally, edge servers solve the following

Authorized licensed use limited to: BRAC UNIVERSITY. Downloaded on February 25,2026 at 08:15:36 UTC from IEEE Xplore. Restrictions apply.

optimization problem:

$$\forall e, \quad f_e^* = \arg \max_{f_e \in [0,1]} U_e(f_e). \tag{27}$$

The first-order necessary condition for optimality is:

$$\frac{\partial U_e}{\partial f_e} = -\frac{\partial C_e}{\partial f_e} - \gamma_e \frac{\partial \epsilon_p}{\partial f_e} = 0. \tag{28}$$

In above equation, $\frac{\partial C_e}{\partial f_e}$ represents the increase in communication cost due to the higher aggregation frequency, while $\frac{\partial \epsilon_p}{\partial f_e}$ accounts for the accuracy degradation resulting from compression at the edge level.

At the highest level of the hierarchy, the central server sets the global policy parameter δ to influence the decision-making process of edge servers. The central server aims to optimize δ to maximize its utility function, anticipating the equilibrium responses of edge servers and clients. The optimization problem for the central server is formulated as:

$$\delta^* = \arg \max_{\delta} U_c(\delta), \tag{29}$$

where $U_c(\delta)$ depends on the aggregated behavior of edge servers, which in turn is influenced by the optimal responses of clients. To ensure that the hierarchical game has a well-defined solution, we analyze the existence of equilibria at different levels:

- *Client-Level and Edge-Level Equilibria:* The concavity of utility functions $U_i(f_i)$ and $U_e(f_e)$ ensures that a unique Nash equilibrium exists for clients and edge servers.
- *Stackelberg Equilibrium:* By applying backward induction, the optimal policy δ^* set by the central server exists under the conditions of continuity and compactness in the strategy space.

To efficiently compute the equilibrium strategies for clients, edge servers, and the central server, we employ a gradient-based iterative update mechanism. Each client updates its communication frequency f_i using a gradient ascent step based on the first-order condition:

$$f_i^{(t+1)} = f_i^{(t)} + \eta_i \left(-\frac{\partial C_i}{\partial f_i} - \alpha_p \frac{\partial \epsilon_p}{\partial f_i} - \alpha_q \frac{\partial \epsilon_q}{\partial f_i} \right). \tag{30}$$

Here, η_i is the step size, and the term in parentheses represents the gradient of the utility function with respect to f_i. Edge servers iteratively adjust their aggregation frequencies based on their respective first-order conditions:

$$f_e^{(t+1)} = f_e^{(t)} + \eta_e \left(-\frac{\partial C_e}{\partial f_e} - \gamma_e \frac{\partial \epsilon_p}{\partial f_e} \right). \tag{31}$$

Here, η_e is the step size for edge-level updates. The central server dynamically refines its policy δ using the following update rule:

$$\delta^{(t+1)} = \delta^{(t)} + \eta_c \left(-\sum_{e=1}^{E} \frac{\partial C_e}{\partial \delta} - \frac{\partial}{\partial \delta} (\delta \cdot \text{Global Error}) \right). \tag{32}$$

This update ensures that the central server optimally balances communication costs and model accuracy. To ensure that the proposed gradient-based updates converge to the Stackelberg equilibrium, we make the assumptions that gradients of U_i, U_e, U_c are assumed to be Lipschitz continuous. The utility functions $U_i(f_i)$ and $U_e(f_e)$ are assumed to be strongly concave to guarantee uniqueness. Algorithm 2 summarizes the working of communication frequency optimization model in our proposed work.

F. Selective Homomorphic Encryption

To enhance privacy in our proposed HFL framework, we introduced Selective Homomorphic Encryption (SHE) motivated from [56], which selectively encrypts critical model updates while allowing less sensitive updates to be transmitted in plaintext. This encryption approach balances privacy, computational efficiency, and communication overhead. By encrypting only a subset of sensitive parameters, the selective design ensures that the added computational overhead on clients remains minimal compared to full-model encryption.

Let $\mathbf{w}_k^t$ denote the local model parameters of client k at communication round t. Instead of encrypting the entire model update, we selectively encrypt *a subset of parameters* based on their importance. Let $\mathcal{I}_k$ represent the index set of parameters selected for encryption, determined by their sensitivity:

$$\mathcal{I}_k = \{m \mid |\nabla F_k(w_{k,m}^t)| > \tau_k\}, \tag{33}$$

where τ_k is a threshold defining the importance of parameter m, based on its gradient magnitude $\nabla F_k(w_{k,m}^t)$. Parameters exceeding τ_k are considered sensitive and are encrypted. For parameters in $\mathcal{I}_k$, we apply HE using a public-key encryption scheme. Let $E(\cdot)$ denote the encryption function. The encrypted model update for client k is:

$$\mathbf{w}_k^E = \{E(w_{k,m}^t) \mid m \in \mathcal{I}_k\} \cup \{w_{k,m}^t \mid m \notin \mathcal{I}_k\}. \tag{34}$$

This ensures that only a fraction of the model parameters is encrypted, reducing computational and transmission overhead. Each edge server e receives encrypted and plaintext updates from its assigned clients $\mathcal{C}_e$. The aggregation at the edge server is performed as:

$$\mathbf{W}_e = \frac{1}{|\mathcal{C}_e|} \sum_{i \in \mathcal{C}_e} \mathbf{w}_i^E. \tag{35}$$

Homomorphic properties allow aggregation of encrypted parameters without decryption, ensuring privacy protection at intermediate nodes. The central server aggregates encrypted updates from all edge servers:

$$\mathbf{W}_g^E = \frac{1}{|\mathcal{E}|} \sum_{e \in \mathcal{E}} \mathbf{W}_e. \tag{36}$$

Decryption is applied selectively to retrieve sensitive parameters:

$$\mathbf{W}_g = \{D(W_{g,m}^E) \mid m \in \mathcal{I}_g\} \cup \{W_{g,m} \mid m \notin \mathcal{I}_g\}, \tag{37}$$

where $D(\cdot)$ represents the decryption function and $\mathcal{I}_g$ is the set of globally encrypted parameters. By encrypting only a subset of model updates, the communication overhead is reduced by a factor of:

$$C_{\text{reduction}} = 1 - \frac{|\mathcal{I}_k|}{d}, \tag{38}$$

Authorized licensed use limited to: BRAC UNIVERSITY. Downloaded on February 25,2026 at 08:15:36 UTC from IEEE Xplore. Restrictions apply.

where d is the total number of parameters. This approach ensures strong privacy guarantees for critical parameters while maintaining communication efficiency.

1) Computational and Communication Overhead of SHE: Our privacy module applies *Selective Homomorphic Encryption (SHE)* only to parameters whose gradient magnitudes exceed a client specific threshold τ_k, forming the encrypted index set I_k. Encryption and decryption are performed on $|I_k|$ coordinates, not all d. Edge and central aggregation operate homomorphically on these encrypted coordinates, while the remaining $(d - |I_k|)$ are aggregated in plaintext. Because $|I_k|/d \ll 1$ in practice, ciphertext expansion affects only a small subset, and pruning and quantization reduce the overall update size. This selectivity preserves privacy of sensitive coordinates with bounded extra compute and keeps net communication below baselines without SHE. We report $|I_k|/d$, client encryption time, server decryption time, and end to end round time in our ablation.

G. Verifiability

In the standard HFL setup, each edge server e aggregates model updates from its assigned clients $\mathcal{C}_e$ and transmits the aggregated model to the central server. To improve verifiability, we introduced an additional step where edge servers share their aggregated models with one another before sending them to the global server. Let $\mathbf{W}_e$ denote the aggregated model at edge server e. Instead of directly transmitting $\mathbf{W}_e$ to the central server, each edge server first shares its model with other edge servers in the network. Let $\mathcal{E}$ be the set of all edge servers. This step allows edge servers to cross-verify their models against the updates of other nodes, mitigating risks associated with poisoned or compromised updates from adversarial clients. After model sharing and aggregation at the edge level, the final edge-verified model is transmitted to the central server. To ensure verifiability, the central server performs an integrity check by comparing individual edge models with the globally aggregated model:

$$\delta_e = \|\mathbf{W}_e^{\text{shared}} - \mathbf{W}_g\|_2. \tag{39}$$

The threshold ϵ is empirically defined as a small fraction of the global model norm, for example $\epsilon = 0.01\|W_g\|$, ensuring that only significant inconsistencies are flagged while avoiding false positives. The value 0.01 was selected empirically as it provided a good balance between sensitivity to malicious or inconsistent updates and minimizing false positives during aggregation. If δ_e exceeds the predefined threshold ϵ, indicating a potential anomaly or malicious activity, the corresponding edge model is flagged for further review. This step ensures that any compromised or inconsistent model updates are detected before being integrated into the final global model. SHE and the Verifiable Edge Model Exchange mechanism were briefly discussed here, and we will extend these ideas in our future research.

IV. Experimental Evaluations and Results

In this section, we evaluated the performance of our proposed HFL framework. The proposed model was compared against baseline methods FedAvg, FedProx and state-of-the-art FL approaches FedLP [23], AQUILA [59], LAQ [58] across various performance metrics and experimental scenarios. The experiments were conducted on four widely used benchmark datasets: MNIST, Fashion-MNIST (FMNIST), CIFAR-10, and CIFAR-100. MNIST and FMNIST consisted of 60,000 training images and 10,000 test images (grayscale, 28×28), with the former containing handwritten digits and the latter featuring images of clothing items. CIFAR-10 and CIFAR-100 comprised 50,000 training images and 10,000 test images (color, 32×32), with CIFAR-10 containing ten object classes and CIFAR-100 featuring a more fine-grained classification with 100 classes. To evaluate the framework under diffet data distribution settings, both IID (Independent and Identically Distributed) and non-IID (Non-Independent and Non-Identically Distributed) scenarios were considered.

For the IID setting, stratified sampling was used to ensure an equal distribution of classes among clients. For the non-IID setting, data heterogeneity was introduced using a Dirichlet distribution to simulate realistic variations in client data, where some clients had access to only a subset of the classes.

The model used for evaluation was a Convolutional Neural Network (CNN) optimized for FL, consisting of multiple convolutional layers followed by batch normalization, ReLU activation, and fully connected layers. The learning rate was set to 0.01 with a decay factor, and the Adam optimizer was used for efficient parameter updates. The activation function employed in hidden layers was ReLU, while the output layer used softmax for classification. In addition, the cross-entropy loss function was used for optimization. Training was performed over 200 communication rounds, with each client performing local updates for 5 epochs before aggregation.

In Fig. 4, we evaluate the accuracy performance of our proposed framework and compare it with baseline methods FedAvg and FedProx, along with state-of-the-art approaches: [23], [58], [59]. The y-axis represents accuracy, while the x-axis denotes the number of communication rounds. Fig. 4(a) presents the results on the MNIST dataset, demonstrating that our proposed framework consistently outperforms all comparative schemes in terms of accuracy. As communication rounds progress, accuracy increases for all methods. However, our proposed framework exhibits significantly faster convergence. LAQ remains the second-best performing scheme, followed by FedLP and AQUILA, while FedProx shows better results than FedAvg. The effectiveness of our communication-efficient HFL framework, combined with model communication and aggregation frequency optimization, reinforces its superiority in both accuracy and convergence speed. Fig. 4(b) presents the results on the FMNIST dataset, demonstrating that our proposed framework consistently outperforms all comparative schemes in terms of accuracy. As communication rounds progress, accuracy increases for all methods. However, our proposed framework exhibits significantly faster convergence. LAQ remains the second-best performing scheme, followed by FedLP and AQUILA, while FedProx shows better results than FedAvg. The effectiveness of our communication-efficient HFL framework, combined with communication frequency

Authorized licensed use limited to: BRAC UNIVERSITY. Downloaded on February 25,2026 at 08:15:36 UTC from IEEE Xplore. Restrictions apply.

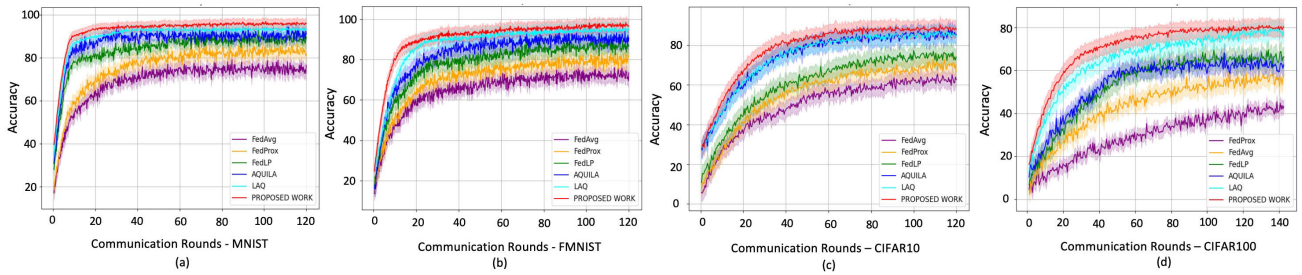


Fig. 4. Accuracy performance comparison of the proposed work with comparative schemes over communication rounds on (a) MNIST, (b) FMNIST, (c) CIFAR-10, and (d) CIFAR-100 datasets.

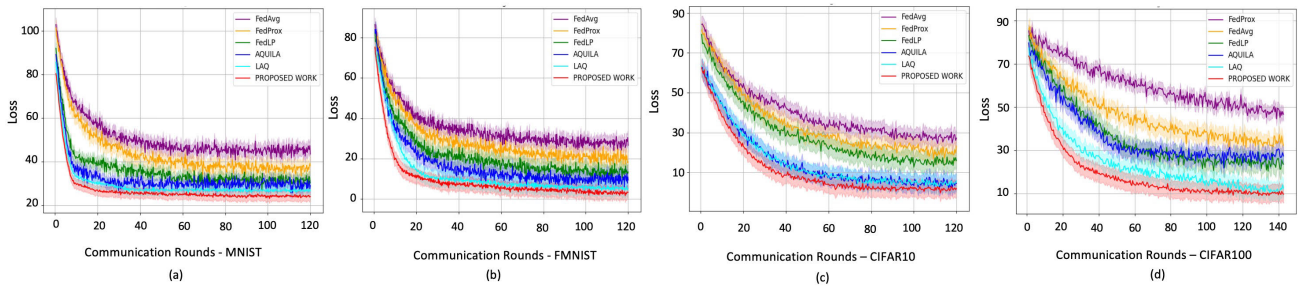


Fig. 5. Loss performance comparison of the proposed work with comparative schemes over communication rounds on (a) MNIST, (b) FMNIST, (c) CIFAR-10, and (d) CIFAR-100 datasets.

optimization, reinforces its superiority in both accuracy and convergence speed.

Fig. 4(c) and Fig. 4(d) present the results on CIFAR-10 and CIFAR-100, which are more complex datasets compared to MNIST and FMNIST due to their larger feature space. The results demonstrate that our proposed framework significantly outperforms all comparative schemes in terms of accuracy. As communication rounds progress, accuracy improves for all methods, but the gap between our framework and other approaches remains substantial. Additionally, our framework exhibits much faster convergence, especially on these more challenging datasets. These findings highlight the effectiveness of our communication-efficient HFL framework, where communication frequency optimization enhances both accuracy and convergence speed, even in complex scenarios. Furthermore, our framework consistently outperforms all comparative schemes on both CIFAR-10 and CIFAR-100. Notably, the accuracy difference between CIFAR-10 and CIFAR-100 is much smaller in our proposed framework compared to others, demonstrating its robustness and adaptability across varying dataset complexities.

Fig. 5 illustrates the loss performance of our proposed framework compared to existing approaches across four datasets: MNIST (a), FMNIST (b), CIFAR-10 (c), and CIFAR-100 (d). The y-axis represents the loss, while the x-axis denotes the number of communication rounds. Across all datasets, our proposed framework consistently achieves the lowest loss, demonstrating superior optimization efficiency. In Fig. 5(a), which presents results for the MNIST dataset, our framework rapidly reduces loss within the initial communication rounds and stabilizes at a significantly lower value than other approaches. A similar trend is observed in Fig. 5(b) for FMNIST, where our method outperforms others by maintaining lower loss throughout the training process. These results confirm the efficiency of our approach in handling

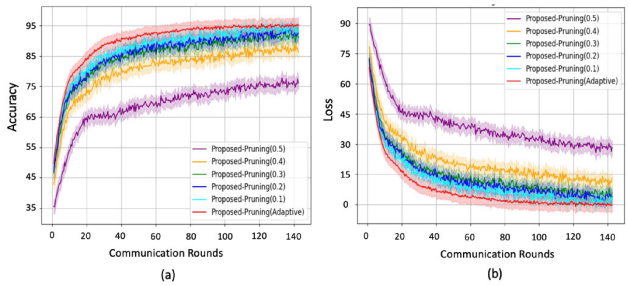


Fig. 6. Analysis of (a) accuracy and (b) loss of the proposed work over different communication rounds and various pruning ratios on the MNIST dataset.

relatively simple datasets. On the more complex CIFAR-10 in Fig. 5(c) and CIFAR-100 Fig. 5(d) datasets, our framework maintains its advantage, achieving lower loss values at a much faster rate. The performance gap between our framework and other comparative schemes is more pronounced, particularly in CIFAR-100, where the reduction in loss is significantly quicker and more stable.

Fig. 6 presents the performance of our proposed framework with different pruning ratios, evaluating both accuracy in Fig. 6(a) and loss in Fig. 6(b) across communication rounds. The results demonstrate that pruning plays a significant role in performance, with lower pruning ratios achieving better accuracy and lower loss. Among all configurations, the adaptive pruning approach consistently outperforms fixed pruning ratios, achieving the highest accuracy and the lowest loss. In Fig 6(a), accuracy improves for all pruning ratios as communication rounds increase, but the adaptive pruning approach achieves the highest accuracy, stabilizing above all other configurations. In contrast, higher pruning ratios (e.g., 0.5) lead to lower final accuracy, indicating the trade-off between model sparsity and performance. Fig. 1(b) shows the corresponding loss curves, where the adaptive pruning approach achieves the

Authorized licensed use limited to: BRAC UNIVERSITY. Downloaded on February 25,2026 at 08:15:36 UTC from IEEE Xplore. Restrictions apply.

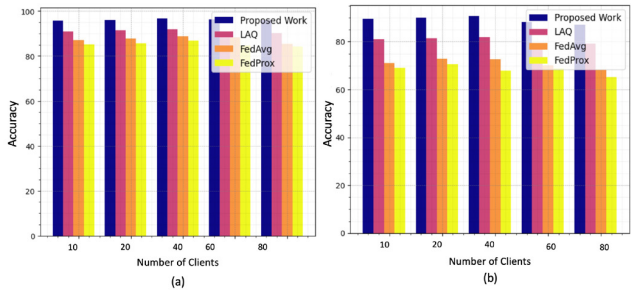


Fig. 7. Analysis of accuracy over different numbers of clients for the proposed and comparative schemes on (a) MNIST and (b) CIFAR-10 datasets.

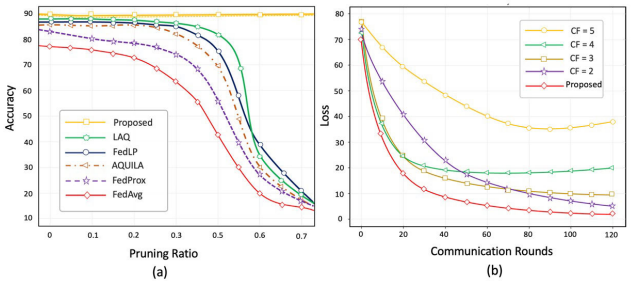


Fig. 8. Analysis of (a) Accuracy of the proposed work compared to comparative schemes with various pruning ratios on the FMNIST dataset, and (b) Loss of the proposed work with different aggregation frequencies.

fastest and most stable convergence, maintaining the lowest loss values throughout training. In comparison, higher pruning ratios exhibit significantly higher loss, indicating slower convergence and reduced model performance. Adaptive pruning in our proposed framework optimally balances model sparsity and performance, leading to superior accuracy and faster convergence compared to fixed pruning strategies.

Fig. 7 illustrates the accuracy performance of our proposed framework under varying numbers of clients, comparing it with existing approaches. Fig. 7(a) presents results on the MNIST dataset, while Fig. 7(b) shows performance on the more complex CIFAR-10 dataset. The x-axis represents the number of participating clients, while the y-axis denotes accuracy. Across all settings, our proposed framework consistently achieves the highest accuracy, demonstrating its robustness in different client environments. In Fig. 7(a), on the MNIST dataset, accuracy remains high across all client variations, with our proposed framework significantly outperforming other methods. Even as the number of clients increases from 10 to 80, our method maintains superior performance, showing its scalability and efficiency in FL settings. Other approaches experience a slight decline in accuracy with increasing clients, indicating a performance gap due to communication and aggregation challenges. Fig. 7(b) shows the results on CIFAR-10, a more complex dataset. Similar to MNIST, our framework consistently delivers the highest accuracy across different client numbers. However, the gap between our method and others is even more pronounced, demonstrating its effectiveness in handling higher-dimensional data with diverse client distributions. While competing methods show a notable decrease in accuracy as the number of clients grows, our approach remains stable, proving its adaptability to varying client participation scenarios.

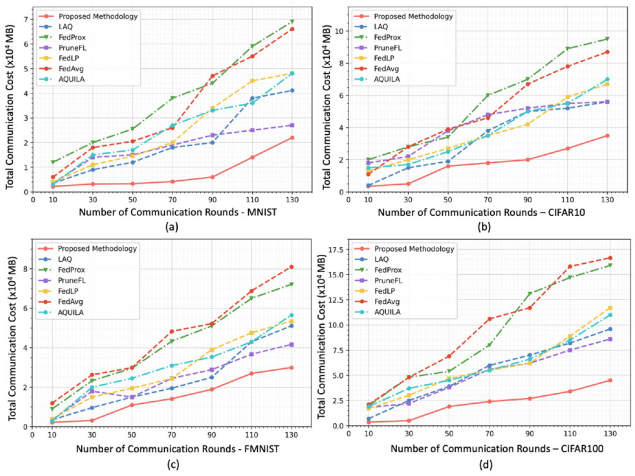


Fig. 9. Analysis of total communication overhead of proposed methodology and comparative schemes over (a) MNIST (b) CIFAR10 (c) FMNIST and (d) CIFAR100 datasets.

Fig. 8 presents the accuracy performance of our proposed framework under two different pruning ratios in Fig. 8(a) and varying communication frequencies in Fig. 8(b). The results demonstrate the robustness of our approach in consistency in better accuracy and lower loss across different configurations. In Fig. 8(a), the impact of different pruning ratios on accuracy is evaluated. As the pruning ratio increases, the accuracy initially remains stable but starts to decline significantly beyond a certain threshold. Our proposed framework achieves a stable and better accuracy across all pruning ratios, outperforming other comparative schemes. Notably, competing methods exhibit lower accuracy values, especially at higher pruning ratios, indicating a decline in model performance. These results confirm the effectiveness of our framework in handling network sparsity while maintaining optimization efficiency. Fig. 8(b) examines the effect of varying communication frequencies (CF) on loss reduction. Our proposed framework is tested under different CF values ranging from 2 to 5. The results show that lower CF values lead to significantly faster convergence and lower loss, while higher CF values result in slower convergence and higher final loss. This highlights the importance of optimizing communication frequency to achieve better trade-offs between efficiency and model performance. Our proposed method, which leverages a Stackelberg game in edge environments, dynamically optimizes model communication frequency and aggregation frequency, achieves faster convergence, and significantly reduces model loss compared to constant-frequency approaches. Our method demonstrates stable and effective learning across all CF configurations, further proving its adaptability in FL environments.

Fig. 9 presents the total communication cost comparison of proposed framework and baseline strategies. The results in Fig. 9(a) on MNIST and Fig. 9(b) CIFAR-10 clearly show that the Proposed Methodology consistently achieves the lowest communication cost across communication rounds compared to existing baselines. Methods such as FedProx and FedAvg depicts significantly higher costs, as they transmit full-precision updates without aggressive compression. LAQ and AQUILA reduce communication through quantization, but

Authorized licensed use limited to: BRAC UNIVERSITY. Downloaded on February 25,2026 at 08:15:36 UTC from IEEE Xplore. Restrictions apply.

their efficiency gains are still limited compared to the proposed approach. PruneFL and FedLP show moderate savings by pruning and low-rank updates. Furthermore, the results in Fig. 9(c) on FMNIST and Fig. 9(d) CIFAR-100 shows that our proposed work outperforms the baseline and comparative schemes. Overall, these findings justify that the proposed method is more communication-efficient, effectively balancing accuracy and cost, which is crucial for resource-constrained FL environments.

V. Conclusion

In this work, we proposed a novel Hierarchical FL framework that optimizes communication efficiency while maintaining high model accuracy and privacy. Our approach integrates dynamic model pruning and quantization, allowing adaptive adjustments to pruning ratios and quantization levels, significantly reducing communication overhead without compromising performance. Additionally, we introduced a Stackelberg game-based model communication and aggregation frequency optimization mechanism, where clients, edge servers, and the central server collaboratively determine optimal update frequencies, ensuring efficient convergence with minimal bandwidth consumption. To enhance privacy and trust, we incorporated Selective Homomorphic Encryption and a verifiable model trust assessment, enabling secure computations and reliable participation of edge devices in the FL process. Extensive experimental evaluations demonstrated that our framework outperforms state-of-the-art methods. For future research, we aim to explore adaptive aggregation strategies for further optimizing hierarchical communication in FL and investigate energy-efficient FL to reduce computational burdens on resource-constrained edge devices. Moreover, we plan to extend our framework to address adversarial threats such as model poisoning and inference attacks, thereby further enhancing the security and resilience of FL systems.

References

[1] D. C. Nguyen, M. Ding, P. N. Pathirana, A. Seneviratne, J. Li, and H. Vincent Poor, "Federated learning for Internet of Things: A comprehensive survey," *IEEE Commun. Surveys Tuts.*, vol. 23, no. 3, pp. 1622–1658, 3rd Quart., 2021.

[2] H. Liu, F. He, and G. Cao, "Communication-efficient federated learning for heterogeneous edge devices based on adaptive gradient quantization," in *Proc. IEEE Conf. Comput. Commun. (INFOCOM)*, May 2023, pp. 1–10.

[3] Y. Jia et al., "DapperFL: Domain adaptive federated learning with model fusion pruning for edge devices," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 37, Dec. 2024, pp. 13099–13123.

[4] X. Han et al., "Energy-efficient wireless federated learning via doubly adaptive quantization," *IEEE Trans. Cognit. Commun. Netw.*, vol. 11, no. 4, pp. 2216–2231, Aug. 2025.

[5] A. Tariq, T. Qayyum, M. A. Serhani, F. M. Sallabi, I. Taleb, and E. S. Barka, "Enhancing communication efficiency in fl with adaptive gradient quantization and communication frequency optimization," in *Proc. IEEE Int. Conf. Commun.*, Jun. 2025, pp. 1201–1206.

[6] X. Chen, A. Wang, X. Deng, and J. Gui, "Latency-efficient wireless federated learning with sparsification and quantization for heterogeneous devices," *IEEE Internet Things J.*, vol. 12, no. 1, pp. 488–501, Jan. 2025.

[7] G. Wang, C. Zhao, Q. Qi, R. Han, L. Bai, and J. Choi, "Efficient federated learning via joint communication and computation optimization," *IEEE Trans. Veh. Technol.*, vol. 73, no. 8, pp. 11580–11592, Aug. 2024.

[8] A. Tariq et al., "Trustworthy federated learning: A comprehensive review, architecture, key challenges, and future research prospects," *IEEE Open J. Commun. Soc.*, vol. 5, pp. 4920–4998, 2024, doi: 10.1109/OJCOMS.2024.3438264.

[9] H. Cheng, M. Zhang, and J. Q. Shi, "A survey on deep neural network pruning: Taxonomy, comparison, analysis, and recommendations," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 10558–10578, Dec. 2024.

[10] Y. Jiang et al., "Model pruning enables efficient federated learning on edge devices," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 12, pp. 10374–10386, Dec. 2023.

[11] Z. Chen, W. Yi, H. Shin, and A. Nallanathan, "Adaptive model pruning for communication and computation efficient wireless federated learning," *IEEE Trans. Wireless Commun.*, vol. 23, no. 7, pp. 7582–7598, Jul. 2024.

[12] S. Liu, G. Yu, R. Yin, J. Yuan, L. Shen, and C. Liu, "Joint model pruning and device selection for communication-efficient federated edge learning," *IEEE Trans. Commun.*, vol. 70, no. 1, pp. 231–244, Jan. 2022.

[13] X. Chang, M. S. Obaidat, J. Ma, X. Xue, Y. Yu, and X. Wu, "Efficient federated learning via adaptive model pruning for Internet of Vehicles with a constrained latency," *IEEE Trans. Sustain. Comput.*, vol. 10, no. 2, pp. 300–316, Mar. 2025, doi: 10.1109/TSUSC.2024.3441658.

[14] L. Yi, X. Shi, N. Wang, J. Zhang, G. Wang, and X. Liu, "FedPE: Adaptive model pruning-expanding for federated learning on mobile devices," *IEEE Trans. Mobile Comput.*, vol. 23, no. 11, pp. 10475–10493, Nov. 2024.

[15] Z. Jiang et al., "Computation and communication efficient federated learning with adaptive model pruning," *IEEE Trans. Mobile Comput.*, vol. 23, no. 3, pp. 2003–2021, Mar. 2024.

[16] G. Zhou et al., "FedPAGE: Pruning adaptively toward global efficiency of heterogeneous federated learning," *IEEE/ACM Trans. Netw.*, vol. 32, no. 3, pp. 1873–1887, Jun. 2024, doi: 10.1109/TNET.2023.3328632.

[17] Y. Xu et al., "Enhancing decentralized federated learning with model pruning and adaptive communication," *IEEE Trans. Ind. Informat.*, vol. 21, no. 1, pp. 70–84, Jan. 2025.

[18] X. Liu, T. Ratnarajah, M. Sellathurai, and Y. C. Eldar, "Adaptive model pruning and personalization for federated learning over wireless networks," *IEEE Trans. Signal Process.*, vol. 72, pp. 4395–4411, 2024, doi: 10.1109/TSP.2024.3459808.

[19] J. Pei, W. Li, and S. Mumtaz, "From routine to reflection: Pruning neural networks in communication-efficient federated learning," *IEEE Trans. Artif. Intell.*, early access, Sep. 17, 2025, doi: 10.1109/TAI.2024.3462300.

[20] H. Liu et al., "FedADP: Communication-efficient by model pruning for federated learning," in *Proc. GLOBECOM - IEEE Global Commun. Conf.*, Dec. 2023, pp. 3093–3098.

[21] Y. Shi et al., "Efficient federated learning with enhanced privacy via lottery ticket pruning in edge computing," *IEEE Trans. Mobile Comput.*, vol. 23, no. 10, pp. 9946–9958, Oct. 2024, doi: 10.1109/TMC.2024.3370967.

[22] Q. Li, R. Cai, and Y. Zhu, "GHPPFL: A privacy preserving federated learning based on gradient compression and homomorphic encryption in consumer app security," *IEEE Trans. Consum. Electron.*, vol. 71, no. 2, pp. 5090–5099, May 2025.

[23] Z. Zhu et al., "FedLP: Layer-wise pruning mechanism for communication-computation efficient federated learning," in *Proc. IEEE Int. Conf. Commun.*, May 2023, pp. 1250–1255.

[24] Z. Chen, J. Peng, J. Kang, and D. Niyato, "FedGroup-prune: IoT device amicable and training-efficient federated learning via combined group lasso sparse model pruning," *IEEE Internet Things J.*, vol. 11, no. 24, pp. 40921–40932, Dec. 2024.

[25] X. Wang, H. Yu, Y. Chen, R. O. Sinnott, and Z. Yang, "PrVFL: Pruning-aware verifiable federated learning for heterogeneous edge computing," *IEEE Trans. Mobile Comput.*, vol. 23, no. 12, pp. 15062–15079, Dec. 2024.

[26] P. Prakash et al., "IoT device friendly and communication-efficient federated learning via joint model pruning and quantization," *IEEE Internet Things J.*, vol. 9, no. 15, pp. 13638–13650, Aug. 2022.

[27] X. Liu, S. Wang, Y. Deng, and A. Nallanathan, "Adaptive federated pruning in hierarchical wireless networks," *IEEE Trans. Wireless Commun.*, vol. 23, no. 6, pp. 5985–5999, Jun. 2024.

[28] X. Liu, S. Wang, Y. Deng, and A. Nallanathan, "Adaptive model pruning for hierarchical wireless federated learning," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Apr. 2024, pp. 1–6.

[29] M. F. Pervej, R. Jin, and H. Dai, "Hierarchical federated learning in wireless networks: Pruning tackles bandwidth scarcity and system heterogeneity," *IEEE Trans. Wireless Commun.*, vol. 23, no. 9, pp. 11417–11432, Sep. 2024.

Authorized licensed use limited to: BRAC UNIVERSITY. Downloaded on February 25,2026 at 08:15:36 UTC from IEEE Xplore. Restrictions apply.

[30] Y. Mao et al., "Communication-efficient federated learning with adaptive quantization," *ACM Trans. Intell. Syst. Technol.*, vol. 13, no. 4, pp. 1–26, Aug. 2022.

[31] R. Hönig, Y. Zhao, and R. F. Mullins, "DAdaQuant: Doubly-adaptive quantization for communication-efficient federated learning," in *Proc. ICML*, 2021, pp. 8852–8866.

[32] G. Chen et al., "NQFL: Nonuniform quantization for communication efficient federated learning," *IEEE Commun. Lett.*, vol. 28, no. 2, pp. 332–336, Feb. 2024.

[33] Y. Ren, Y. Cao, C. Ye, and X. Cheng, "Two-layer accumulated quantized compression for communication-efficient federated learning: TLAQC," *Sci. Rep.*, vol. 13, no. 1, p. 11658, Jul. 2023.

[34] Z. Long, Y. Chen, H. Dou, Y. Zhang, and Y. Chen, "FedSQ: Sparse-quantized federated learning for communication efficiency," *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 4050–4061, Feb. 2024.

[35] J. Oh, D. Lee, D. Won, W. Noh, and S. Cho, "Communication-efficient federated learning over-the-air with sparse one-bit quantization," *IEEE Trans. Wireless Commun.*, vol. 23, no. 10, pp. 15673–15689, Oct. 2024.

[36] Y. Liu, P. Huang, F. Yang, K. Huang, and L. Shu, "QuAsyncFL: Asynchronous federated learning with quantization for cloud–edge–terminal collaboration enabled AIoT," *IEEE Internet Things J.*, vol. 11, no. 1, pp. 59–69, Jan. 2024.

[37] X. Lyu et al., "Secure and efficient federated learning with provable performance guarantees via stochastic quantization," *IEEE Trans. Inf. Forensics Security*, vol. 19, pp. 4070–4085, 2024.

[38] X. Hou, J. Wang, C. Jiang, Z. Meng, J. Chen, and Y. Ren, "Efficient federated learning for metaverse via dynamic user selection, gradient quantization and resource allocation," *IEEE J. Sel. Areas Commun.*, vol. 42, no. 4, pp. 850–866, Apr. 2024.

[39] D. Jhunjhunwala, A. Gadhikar, G. Joshi, and Y. C. Eldar, "Adaptive quantization of model updates for communication-efficient federated learning," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Jun. 2021, pp. 3110–3114.

[40] J. Wang and G. Joshi, "Adaptive communication strategies to achieve the best error-runtime trade-off in local-update SGD," in *Proc. Mach. Learn. Syst.*, 2018, pp. 212–229.

[41] L. Qu, S. Song, and C.-Y. Tsui, "FedDQ: Communication-efficient federated learning with descending quantization," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, Dec. 2022, pp. 281–286.

[42] P. S. Bouzinis, P. D. Diamantoulakis, and G. K. Karagiannidis, "Wireless quantized federated learning: A joint computation and communication design," *IEEE Trans. Commun.*, vol. 71, no. 5, pp. 2756–2770, May 2023.

[43] H. Chen and H. Vikalo, "Mixed-precision quantization for federated learning on resource-constrained heterogeneous devices," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 6138–6148.

[44] J. Xu, W. Du, Y. Jin, W. He, and R. Cheng, "Ternary compression for communication-efficient federated learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 3, pp. 1162–1176, Mar. 2022.

[45] Y. Ji and L. Chen, "FedQNN: A computation–communication-efficient federated learning framework for IoT with low-bitwidth neural network quantization," *IEEE Internet Things J.*, vol. 10, no. 3, pp. 2494–2507, Feb. 2023.

[46] L. Chen, W. Liu, Y. Chen, and W. Wang, "Communication-efficient design for quantized decentralized federated learning," *IEEE Trans. Signal Process.*, vol. 72, pp. 1175–1188, 2024.

[47] L. Liu, J. Zhang, S. Song, and K. B. Letaief, "Hierarchical federated learning with quantization: Convergence analysis and system design," *IEEE Trans. Wireless Commun.*, vol. 22, no. 1, pp. 2–18, Jan. 2023.

[48] S. Chen, C. Shen, L. Zhang, and Y. Tang, "Dynamic aggregation for heterogeneous quantization in federated learning," *IEEE Trans. Wireless Commun.*, vol. 20, no. 10, pp. 6804–6819, Oct. 2021.

[49] Q. Lin, S. Jiang, Z. Zhen, T. Chen, C. Wei, and H. Lin, "Fed-PEMC: A privacy-enhanced federated deep learning algorithm for consumer electronics in mobile edge computing," *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 4073–4086, Feb. 2024.

[50] A. Tariq, F. Sallabi, M. A. Serhani, T. Qayyum, and E. S. Barka, "Leveraging game theory and XAI for data quality-driven sample and client selection in trustworthy split federated learning," *IEEE Trans. Consum. Electron.*, vol. 71, no. 2, pp. 6686–6699, May 2025.

[51] S. Chen, L. Li, G. Wang, M. Pang, and C. Shen, "Federated learning with heterogeneous quantization bit allocation and aggregation for Internet of Things," *IEEE Internet Things J.*, vol. 11, no. 2, pp. 3132–3143, Jan. 2024.

[52] A. Tariq, F. Sallabi, M. A. Serhani, and E. Baraka, "A trust and data quality-based dynamic node selection and aggregation optimization in federated learning," in *Proc. Int. Wireless Commun. Mobile Comput. (IWCMC)*, May 2024, pp. 1424–1430.

[53] C. Zhang et al., "Distributed deep reinforcement learning-based gradient quantization for federated learning enabled vehicle edge computing," *IEEE Internet Things J.*, vol. 12, no. 5, pp. 4899–4913, Mar. 2025.

[54] Y. Ren, C. Wu, D. K. C. So, and J. Tang, "DRL-based joint aggregation frequency and edge association for energy-efficient hierarchical federated learning," *IEEE Trans. Wireless Commun.*, vol. 24, no. 8, pp. 6841–6856, Aug. 2025.

[55] L. Yang, Y. Gan, J. Cao, and Z. Wang, "Optimizing aggregation frequency for hierarchical model training in heterogeneous edge computing," *IEEE Trans. Mobile Comput.*, vol. 22, no. 7, pp. 4181–4194, Jul. 2023.

[56] A. Tariq, F. M. Sallabi, M. Adel Serhani, T. Qayyum, and E. S. Barka, "Dynamic edge computing with MARL and AI-driven selective homomorphic encryption for trustworthy federated learning," in *Proc. IEEE IEEE Conf. Comput. Commun. Workshops (INFOCOM WKSHPS)*, May 2025, pp. 1–8.

[57] H. Zakerinia, S. Talaei, G. Nadiradze, and D. Alistarh, "Communication-efficient federated learning with data and client heterogeneity," in *Proc. 27th Int. Conf. Artif. Intell. Statist. (AISTATS)*, 2024, pp. 3448–3456.

[58] J. Sun, T. Chen, G. B. Giannakis, Q. Yang, and Z. Yang, "Lazily aggregated quantized gradient innovation for communication-efficient federated learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 4, pp. 2031–2044, Apr. 2022.

[59] Z. Zhao et al., "AQUILA: Communication efficient federated learning with adaptive quantization in device selection strategy," *IEEE Trans. Mobile Comput.*, vol. 23, no. 6, pp. 7363–7376, Jun. 2024.



Asadullah Tariq (Member, IEEE) received the bachelor's degree in computer science from PUCIT, Pakistan, in 2016, and the master's degree in computer science from FAST-NUCES, Pakistan, in 2019. He is currently pursuing the Ph.D. degree in computer science with United Arab Emirates University, United Arab Emirates. He has held an adjunct faculty position with UAE University since 2023. He was a Visiting Student with QMUL, London, U.K., in 2025. He has authored numerous journal articles, conference papers, and book chapters with notable contributions to prestigious journals, such as IEEE COMMUNICATIONS SURVEYS AND TUTORIALS (IF: 35.6), IEEE OPEN JOURNAL OF THE COMMUNICATIONS SOCIETY, IEEE INTERNET OF THINGS JOURNAL, IEEE TRANSACTIONS ON COGNITIVE AND DEVELOPMENTAL SYSTEMS, IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS, IEEE TRANSACTIONS ON CONSUMER ELECTRONICS, IEEE ACCESS, and *Applied Sciences*, and top conferences, such as IEEE INFOCOM, IEEE ICC, ACM/IEEE SEC, VTC-Fall'25, IEEE IWCMC, IEEE FNWF, and EAI SecureComm. He is the author of two published book chapters in Elsevier. His current research interest is developing trustworthy FL frameworks. He got Best Paper Award at IEEE IIT'23 and EAI SecureComm'24, Student travel grant at ACM/IEEE SEC'23, and the Best Presentation Award at BDSIC'23. He was a reviewer for reputed journals, including but not limited to IEEE COMMUNICATIONS SURVEYS AND TUTORIALS, IEEE COMMUNICATIONS MAGAZINE, IEEE TRANSACTIONS ON MOBILE COMPUTING, IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, IEEE TRANSACTIONS ON INDUSTRIAL INFORMATICS, IEEE COMMUNICATIONS LETTERS, IEEE TRANSACTIONS ON ARTIFICIAL INTELLIGENCE, IEEE JOURNAL OF BIOMEDICAL AND HEALTH INFORMATICS, IEEE TRANSACTIONS ON CLOUD COMPUTING, IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS, *JNCA*, IEEE IOT JOURNAL, IEEE ACCESS, *Elsevier Information Sciences*, and *FGCS*.

Authorized licensed use limited to: BRAC UNIVERSITY. Downloaded on February 25,2026 at 08:15:36 UTC from IEEE Xplore. Restrictions apply.



Farag Sallabi received the Ph.D. degree in electrical and computer engineering from the University of Ottawa, Canada, in 2001. He is an Associate Professor with the Department of Computer and Network Engineering and the Coordinator of the Ph.D. Program of informatics and computing with CIT, United Arab Emirates University (UAEU). He was with Sigpro Wireless Company, Ottawa, Canada, from August 2001 to August 2002. He was with UAEU, from September 2002 to June 2012. From July 2012 to July 2015, he was the Director General of the Directorate General of Electronic Services, Ministry of Communications and Informatics, Libya. He has led a project to develop and implement an e-Government program in Libya. He then rejoined the UAEU, in August 2015. He served as the PI and the co-PI on several research projects funded by UAEU. He has published over 70 refereed publications, including journals and conferences. His research interests include quality of service provisioning in wired and wireless networks, routing in wireless sensor networks, performance evaluation of WSN, network management, modeling and simulation, the IoT, and smart healthcare systems.



Mohamed Adel Serhani received the B.Sc. degree in computer science from ESCI, Morocco, in 1996, the M.Sc. degree in software engineering from the University of Montreal, Canada, in 2002, and the Ph.D. degree in computer engineering from Concordia University, in 2006. He is currently a Professor with the College of Computing and Informatics, University of Sharjah. Before joining UoS, he was an Assistant Dean for Research and Graduate Studies with the College of Information Technology, UAE University, from 2018 to 2022, the Ph.D. Program Director from 2018 to 2020, and the Master Program Director from 2014 to 2017. He has a large experience earned throughout his involvement and management of different research and development projects. He has published around 170 refereed publications, including journals, conferences, a book, and book chapters. His research interests include federated learning, cloud for data intensive e-health applications, and services; SLA enforcement in cloud data centers, and big data value chain; cloud federation and monitoring, and non-invasive smart health monitoring; management of communities of web services; and web services applications and security. He has served as an Organizer and the Technical Program Chair/Co-Chair for many international conferences, and workshops, such as IWCMC'23, BDSIC'23, SERVICES'21, ICWS'20, AI-PA'2020, IIT'13, NDT'12, and WMCS'12. He was granted several awards and recognitions including the CIT merit allowance for four times, the CIT Excellence Award in Research, in 2015, the CIT Excellence Award in Teaching, in 2016, and the CIT Excellence Award in Service, in 2019.



Ezedin S. Barka received the Ph.D. degree in information technology from George Mason University, Fairfax, VA, USA, in 2002. He is currently an Associate Professor and the Department Chair with the College of Information Technology, United Arab Emirates University. He was a member of the Laboratory for Information Security Technology (LIST), George Mason University. He has published over 50 journals and conference papers. His current research interests include access control, where he published a number of papers addressing delegation of rights using RBAC. Other research interests include digital rights management (DRM), large-scale security architectures and models, trust management, security in UAVs, and network wired and wireless and distributed systems security. He is a member of the IEEE Communications Society and the IEEE Communications and Information Security Technical Committee (CISTC). He was a reviewer of several international journals and conferences. He serves on the technical program committees for many international IEEE conferences, such as ACSAC, GLOBECOM, ICC, WIMOB, and WCNC.



Tariq Qayyum (Member, IEEE) received the master's degree in information technology from the National University of Science and Technology. He is currently pursuing the Ph.D. degree with the College of Information Technology, United Arab Emirates University (UAEU). He has experience working as a Research Assistant on the project cyber infrastructure protection and malware analysis (CIPMA) with Pakistan Institute of Engineering and Applied Sciences. This project was funded by the Higher Education Commission of Pakistan (HEC) and the National Center for Cyber Security (NCCS), Pakistan. He has authored numerous journal articles, conference papers and book chapters with notable contributions to prestigious journals such as IEEE INTERNET OF THINGS JOURNAL, IEEE TRANSACTIONS ON SUSTAINABLE COMPUTING, IEEE ACCESS, IEEE OPEN JOURNAL OF THE COMMUNICATIONS SOCIETY, and *SN Cloud Computing*, and top conferences such as IEEE INFOCOM, IEEE ICC, ACM/IEEE SEC, IEEE IWCMC, IEEE FNWF, and EAI SecureComm. His research interests include federated learning, network security, cloud computing, fog computing, the Internet of Things (IoT), and distributed systems.



Irfanud Din received the Ph.D. degree from Incheon University, South Korea. He was an Assistant Professor with Inha University, Tashkent, Uzbekistan, and a Principal Scientist with the Center for Research in Distributed and Supercomputing. He was associated with Superior University Lahore as an Associate Professor from 2018 to 2023, and the HoD of the Department of Computer Science. Currently, he is a Professor and the HoD of the Department of Computer Science, New Uzbekistan University, Tashkent. He is also working on various research and development projects in the domain of wireless communication and its integration with machine learning and federated learning.



Ikram Syed received the B.Sc. degree in computer system engineering from NWFP, UET, Peshawar, Pakistan, in 2008, the M.S. degree from Incheon National University, Incheon, South Korea, in 2013, and the Ph.D. degree from the Department of Computer Engineering, Ajou University, Suwon, South Korea, in 2017. He is currently an Associate Professor with the Department of Information and Communication Engineering, Hankuk University of Foreign Studies, South Korea. His research interests include wireless networks, cognitive radio communications, the Internet of Things, and wireless sensor networks.

Authorized licensed use limited to: BRAC UNIVERSITY. Downloaded on February 25,2026 at 08:15:36 UTC from IEEE Xplore. Restrictions apply.